

Questions

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1. (a) Show that an appropriate integrating factor for

$$x^2 \frac{dy}{dx} + y = 1, \quad x > 0$$

is $e^{-1/x}$

[4]

- (b) Hence find the solution of the differential equation

$$x^2 \frac{dy}{dx} + y = 1, \quad x > 0$$

for which $y = 3$ when $x = 1$

[6]

Solution

- (a) First write the differential equation in the standard linear form by dividing through by x^2 :

$$\frac{dy}{dx} + \frac{1}{x^2}y = \frac{1}{x^2}$$

This is of the form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

with

$$P(x) = \frac{1}{x^2} = x^{-2}$$

For a linear differential equation, an integrating factor is

$$\mu(x) = e^{\int P(x) dx}$$

So

$$\begin{aligned} \mu(x) &= e^{\int x^{-2} dx} \\ &= e^{-x^{-1}} \\ &= e^{-1/x} \end{aligned}$$

Hence an appropriate integrating factor is $e^{-1/x}$.

- (b) Using the integrating factor from part (a), multiply

$$\frac{dy}{dx} + \frac{1}{x^2}y = \frac{1}{x^2}$$

through by $e^{-1/x}$:

$$e^{-1/x} \frac{dy}{dx} + \frac{e^{-1/x}}{x^2} y = \frac{e^{-1/x}}{x^2}$$

Now apply the product rule. Since

$$\frac{d}{dx} (e^{-1/x}) = e^{-1/x} \cdot \frac{1}{x^2}$$

we get

$$\begin{aligned} \frac{d}{dx} (ye^{-1/x}) &= e^{-1/x} \frac{dy}{dx} + y \frac{d}{dx} (e^{-1/x}) \\ &= e^{-1/x} \frac{dy}{dx} + \frac{e^{-1/x}}{x^2} y \end{aligned}$$

So the differential equation becomes

$$\frac{d}{dx} (ye^{-1/x}) = \frac{e^{-1/x}}{x^2}$$

Integrating both sides with respect to x gives

$$ye^{-1/x} = \int \frac{e^{-1/x}}{x^2} dx + C$$

But

$$\frac{d}{dx} \left(e^{-1/x} \right) = \frac{e^{-1/x}}{x^2}$$

so

$$\int \frac{e^{-1/x}}{x^2} dx = e^{-1/x}$$

Hence

$$ye^{-1/x} = e^{-1/x} + C$$

Multiplying through by $e^{1/x}$:

$$y = 1 + Ce^{1/x}$$

Now use the condition $y = 3$ when $x = 1$:

$$3 = 1 + Ce$$

$$2 = Ce$$

$$C = \frac{2}{e}$$

Substitute back:

$$\begin{aligned} y &= 1 + \frac{2}{e}e^{1/x} \\ &= 1 + 2e^{1/x-1} \end{aligned}$$

2. (a) Show that

$$\frac{d}{dx}(\ln(\cosh x + 1)) = \tanh\left(\frac{x}{2}\right) \quad [3]$$

(b) Hence solve the differential equation

$$\frac{dy}{dx} + \tanh\left(\frac{x}{2}\right)y = \sinh x$$

for which $y = 2$ when $x = 0$. Give your answer in exact form [7]

Solution

(a) Using $\frac{d}{dx} \ln u = \frac{1}{u} \frac{du}{dx}$,

$$\frac{d}{dx}(\ln(\cosh x + 1)) = \frac{1}{\cosh x + 1} \cdot \frac{d}{dx}(\cosh x + 1) = \frac{\sinh x}{\cosh x + 1}$$

Now use the half-angle identities

$$\sinh x = 2 \sinh\left(\frac{x}{2}\right) \cosh\left(\frac{x}{2}\right), \quad \cosh x + 1 = 2 \cosh^2\left(\frac{x}{2}\right)$$

So

$$\begin{aligned} \frac{\sinh x}{\cosh x + 1} &= \frac{2 \sinh\left(\frac{x}{2}\right) \cosh\left(\frac{x}{2}\right)}{2 \cosh^2\left(\frac{x}{2}\right)} \\ &= \frac{\sinh\left(\frac{x}{2}\right)}{\cosh\left(\frac{x}{2}\right)} \\ &= \tanh\left(\frac{x}{2}\right) \end{aligned}$$

Hence

$$\frac{d}{dx}(\ln(\cosh x + 1)) = \tanh\left(\frac{x}{2}\right)$$

as required.

(b) The differential equation is linear:

$$\frac{dy}{dx} + \tanh\left(\frac{x}{2}\right)y = \sinh x$$

From part (a),

$$\frac{d}{dx}(\ln(\cosh x + 1)) = \tanh\left(\frac{x}{2}\right)$$

so an integrating factor is

$$\mu(x) = e^{\int \tanh\left(\frac{x}{2}\right) dx} = e^{\ln(\cosh x + 1)} = \cosh x + 1$$

Multiplying the differential equation throughout by $\cosh x + 1$,

$$(\cosh x + 1) \frac{dy}{dx} + (\cosh x + 1) \tanh\left(\frac{x}{2}\right)y = (\cosh x + 1) \sinh x$$

The left-hand side is

$$\frac{d}{dx}((\cosh x + 1)y)$$

Hence the differential equation can be written

$$\frac{d}{dx}((\cosh x + 1)y) = (\cosh x + 1) \sinh x$$

We can integrate the right hand side by reversing the chain rule:

$$\int (\cosh x + 1) \sinh x \, dx = \frac{1}{2}(\cosh x + 1)^2 + C$$

So the differential equation becomes

$$\begin{aligned} (\cosh x + 1)y &= \int (\cosh x + 1) \sinh x \, dx \\ &= \frac{1}{2}(\cosh x + 1)^2 + C \end{aligned}$$

Therefore

$$y = \frac{1}{2}(\cosh x + 1) + \frac{C}{\cosh x + 1}$$

Use the condition $y = 2$ when $x = 0$. Since $\cosh 0 = 1$,

$$\begin{aligned} 2 &= \frac{1}{2}(1 + 1) + \frac{C}{1 + 1} \\ 2 &= 1 + \frac{C}{2} \end{aligned}$$

Hence

$$C = 2$$

Substituting back,

$$y = \frac{\cosh x + 1}{2} + \frac{2}{\cosh x + 1}$$

Alternatively, one can integrate using the double angle formula for $\sinh 2x$:

$$\int (\cosh x + 1) \sinh x \, dx = \int (\cosh x \sinh x + \sinh x) \, dx = \int \left(\frac{1}{2} \sinh 2x + \sinh x \right) \, dx$$

to obtain an equivalent form of the final answer:

$$y = \frac{\frac{1}{4} \cosh 2x + \cosh x + \frac{11}{4}}{\cosh x + 1}$$

3. (a) For $x > -1$, determine the general solution of the differential equation

$$(1+x) \frac{dy}{dx} + y = \ln(1+x)$$

giving your answer in the form $y = f(x)$ [4]

- (b) Given that $y = -1$ when $x = 0$, determine the positive value of x for which $y = 0$ [2]

Solution

- (a) For $x > -1$, divide the differential equation by $1+x$:

$$\frac{dy}{dx} + \frac{1}{1+x}y = \frac{\ln(1+x)}{1+x}$$

This is a linear differential equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

with

$$P(x) = \frac{1}{1+x}$$

The integrating factor is

$$\begin{aligned}\mu(x) &= e^{\int \frac{1}{1+x} dx} \\ &= e^{\ln(1+x)}\end{aligned}$$

Since $x > -1$, we have $1+x > 0$, so

$$\mu(x) = 1+x$$

Multiplying the differential equation by $1+x$ gives

$$(1+x) \frac{dy}{dx} + y = \ln(1+x)$$

The left-hand side is the derivative of $(1+x)y$:

$$\frac{d}{dx}((1+x)y) = \ln(1+x)$$

Integrating both sides with respect to x :

$$(1+x)y = \int \ln(1+x) dx + C$$

Now evaluate the integral by letting $u = 1+x$, so $du = dx$:

$$\begin{aligned}\int \ln(1+x) dx &= \int \ln u du \\ &= u \ln u - u + C \\ &= (1+x) \ln(1+x) - (1+x) + C\end{aligned}$$

So

$$(1+x)y = (1+x) \ln(1+x) - (1+x) + C$$

Dividing by $1+x$, which is positive since $x > -1$, gives the general solution

$$y = \ln(1+x) - 1 + \frac{C}{1+x}$$

(b) Use the condition $y = -1$ when $x = 0$.

Substituting into the general solution:

$$-1 = \ln(1 + 0) - 1 + \frac{C}{1 + 0}$$

$$-1 = 0 - 1 + C$$

$$-1 = -1 + C$$

So

$$C = 0$$

Hence

$$y = \ln(1 + x) - 1$$

Now set $y = 0$:

$$0 = \ln(1 + x) - 1$$

$$\ln(1 + x) = 1$$

Exponentiating both sides:

$$1 + x = e$$

$$x = e - 1$$

Since the question asks for the positive value,

$$x = e - 1 \approx 1.72$$

4. (a) Show that, for $0 < x < \frac{\pi}{2}$, an appropriate integrating factor for

$$\cos x \frac{dy}{dx} + y \sin x = x$$

is $\sec x$

[4]

- (b) Hence find the solution of

$$\cos x \frac{dy}{dx} + y \sin x = x$$

for $0 < x < \frac{\pi}{2}$, given that $y = 1$ when $x = \frac{\pi}{4}$

[6]

Solution

- (a) First divide the differential equation by $\cos x$:

$$\frac{dy}{dx} + y \tan x = x \sec x$$

This is now in the linear form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

with

$$P(x) = \tan x$$

So an integrating factor is

$$\begin{aligned} \mu(x) &= e^{\int P(x) dx} \\ &= e^{\int \tan x dx} \\ &= e^{\int \frac{\sin x}{\cos x} dx} \\ &= e^{-\ln |\cos x|} \\ &= e^{\ln |\sec x|} \end{aligned}$$

Since $0 < x < \frac{\pi}{2}$, we have $\cos x > 0$, so $|\sec x| = \sec x$. Hence

$$\mu(x) = e^{\ln(\sec x)} = \sec x$$

Therefore an appropriate integrating factor is $\sec x$.

- (b) Using the result from part (a), the equation is

$$\frac{dy}{dx} + y \tan x = x \sec x$$

Multiply through by the integrating factor $\sec x$:

$$\sec x \frac{dy}{dx} + y \sec x \tan x = x \sec^2 x$$

The left-hand side is the derivative of $y \sec x$, so

$$\frac{d}{dx}(y \sec x) = x \sec^2 x$$

Integrating both sides gives

$$y \sec x = \int x \sec^2 x dx + C$$

Integrate by parts with

$$u = x, \quad \frac{dv}{dx} = \sec^2 x$$

so that

$$\frac{du}{dx} = 1, \quad v = \tan x$$

Then

$$\begin{aligned} \int x \sec^2 x \, dx &= x \tan x - \int \tan x \, dx \\ &= x \tan x - (-\ln(\cos x)) \\ &= x \tan x + \ln(\cos x) \end{aligned}$$

Therefore

$$y \sec x = x \tan x + \ln(\cos x) + C$$

Now multiply through by $\cos x$:

$$\begin{aligned} y &= \cos x (x \tan x + \ln(\cos x) + C) \\ &= x \sin x + \cos x \ln(\cos x) + C \cos x \end{aligned}$$

Use the condition $y = 1$ when $x = \frac{\pi}{4}$. Substituting into

$$y \sec x = x \tan x + \ln(\cos x) + C$$

gives

$$\begin{aligned} 1 \cdot \sec \frac{\pi}{4} &= \frac{\pi}{4} \tan \frac{\pi}{4} + \ln \left(\cos \frac{\pi}{4} \right) + C \\ \sqrt{2} &= \frac{\pi}{4} + \ln \left(\frac{\sqrt{2}}{2} \right) + C \\ \sqrt{2} &= \frac{\pi}{4} - \frac{1}{2} \ln 2 + C \end{aligned}$$

So

$$C = \sqrt{2} - \frac{\pi}{4} + \frac{1}{2} \ln 2$$

Substitute this into the expression for y :

$$y = x \sin x + \cos x \left[\ln(\cos x) + \sqrt{2} - \frac{\pi}{4} + \frac{1}{2} \ln 2 \right]$$

5. (a) Show that, for $x > 0$,

$$\frac{d}{dx} \left(\ln \left(\coth \frac{x}{2} \right) \right) = -\operatorname{cosech} x \quad [3]$$

(b) Hence solve the differential equation

$$\frac{dy}{dx} - \operatorname{cosech} x y = \tanh \frac{x}{2}, \quad x > 0$$

for which $y = 1$ when $x = \ln 3$. Give your answer in exact form

[7]

Solution

(a) Let

$$u = \frac{x}{2}$$

Then, by the chain rule,

$$\begin{aligned} \frac{d}{dx} \left(\ln \left(\coth \frac{x}{2} \right) \right) &= \frac{d}{dx} (\ln(\coth u)) \\ &= \frac{1}{\coth u} \cdot \frac{d}{dx} (\coth u) \\ &= \frac{1}{\coth u} \cdot (-\operatorname{cosech}^2 u) \cdot \frac{1}{2} \end{aligned}$$

Now simplify:

$$\begin{aligned} \frac{1}{\coth u} \cdot (-\operatorname{cosech}^2 u) \cdot \frac{1}{2} &= -\frac{1}{2} \cdot \frac{\operatorname{cosech}^2 u}{\coth u} \\ &= -\frac{1}{2} \cdot \frac{1/\sinh^2 u}{\cosh u / \sinh u} \\ &= -\frac{1}{2 \sinh u \cosh u} \end{aligned}$$

Using $\sinh(2u) = 2 \sinh u \cosh u$,

$$\begin{aligned} -\frac{1}{2 \sinh u \cosh u} &= -\frac{1}{\sinh(2u)} \\ &= -\frac{1}{\sinh x} \\ &= -\operatorname{cosech} x \end{aligned}$$

Therefore,

$$\frac{d}{dx} \left(\ln \left(\coth \frac{x}{2} \right) \right) = -\operatorname{cosech} x$$

as required.

(b) The differential equation is

$$\frac{dy}{dx} - \operatorname{cosech} x y = \tanh \frac{x}{2}$$

This is linear, with integrating factor

$$\begin{aligned} \mu(x) &= e^{\int -\operatorname{cosech} x \, dx} \\ &= e^{\ln(\coth \frac{x}{2})} \\ &= \coth \frac{x}{2} \end{aligned}$$

using part (a).

Next, from part (a),

$$\frac{d}{dx} \left(\ln \left(\coth \frac{x}{2} \right) \right) = -\operatorname{cosech} x$$

so

$$\begin{aligned} \frac{d}{dx} \left(\coth \frac{x}{2} \right) &= \coth \frac{x}{2} \cdot \frac{d}{dx} \left(\ln \left(\coth \frac{x}{2} \right) \right) \\ &= -\operatorname{cosech} x \coth \frac{x}{2} \end{aligned}$$

Multiply the differential equation by $\coth \frac{x}{2}$:

$$\coth \frac{x}{2} \frac{dy}{dx} - \operatorname{cosech} x \coth \frac{x}{2} y = \coth \frac{x}{2} \tanh \frac{x}{2}$$

The left-hand side is

$$\frac{d}{dx} \left(y \coth \frac{x}{2} \right)$$

and the right-hand side is 1, since $\coth \frac{x}{2} \tanh \frac{x}{2} = 1$. Hence

$$\frac{d}{dx} \left(y \coth \frac{x}{2} \right) = 1$$

Integrating,

$$y \coth \frac{x}{2} = \int 1 \, dx = x + C$$

So

$$y = (x + C) \tanh \frac{x}{2}$$

Now use $y = 1$ when $x = \ln 3$:

$$1 = (\ln 3 + C) \tanh \frac{\ln 3}{2}$$

Evaluate $\tanh \frac{\ln 3}{2}$:

$$\begin{aligned} \tanh \frac{\ln 3}{2} &= \frac{e^{\ln 3/2} - e^{-\ln 3/2}}{e^{\ln 3/2} + e^{-\ln 3/2}} \\ &= \frac{e^{\ln 3} - 1}{e^{\ln 3} + 1} \\ &= \frac{3 - 1}{3 + 1} \\ &= \frac{1}{2} \end{aligned}$$

Therefore,

$$\begin{aligned} 1 &= (\ln 3 + C) \cdot \frac{1}{2} \\ 2 &= \ln 3 + C \\ C &= 2 - \ln 3 \end{aligned}$$

Substituting back gives

$$y = (x + 2 - \ln 3) \tanh \frac{x}{2}$$

6. (a) Show that an appropriate integrating factor for

$$x(x-1) \frac{dy}{dx} + (x+1)y = x(x-1)^2$$

$$\text{is } \frac{(x-1)^2}{x} \quad [4]$$

(b) Hence find the solution of the differential equation

$$x(x-1) \frac{dy}{dx} + (x+1)y = x(x-1)^2$$

$$\text{for } x > 1, \text{ given that } y = 2 \text{ when } x = 2 \quad [6]$$

Solution

(a) First write the differential equation in linear form by dividing through by $x(x-1)$:

$$\frac{dy}{dx} + \frac{x+1}{x(x-1)}y = x-1$$

For a linear equation

$$\frac{dy}{dx} + P(x)y = Q(x),$$

an integrating factor is

$$\mu(x) = e^{\int P(x) dx}$$

Here

$$P(x) = \frac{x+1}{x(x-1)}$$

Decompose into partial fractions:

$$\begin{aligned} \frac{x+1}{x(x-1)} &= \frac{A}{x} + \frac{B}{x-1} \\ x+1 &= A(x-1) + Bx \\ x+1 &= (A+B)x - A \end{aligned}$$

Equating coefficients gives

$$A+B=1, \quad -A=1$$

so

$$A=-1, \quad B=2$$

Hence

$$\frac{x+1}{x(x-1)} = -\frac{1}{x} + \frac{2}{x-1}$$

Therefore

$$\begin{aligned} \mu(x) &= e^{\int \left(-\frac{1}{x} + \frac{2}{x-1}\right) dx} \\ &= e^{-\ln x + 2\ln(x-1)} \\ &= \frac{(x-1)^2}{x} \end{aligned}$$

Thus an appropriate integrating factor is

$$\mu(x) = \frac{(x-1)^2}{x}$$

(b) Using the integrating factor from part (a),

$$\mu(x) = \frac{(x-1)^2}{x}$$

Multiply the linear form

$$\frac{dy}{dx} + \frac{x+1}{x(x-1)}y = x-1$$

through by $\mu(x)$:

$$\frac{(x-1)^2}{x} \frac{dy}{dx} + \frac{(x-1)^2}{x} \cdot \frac{x+1}{x(x-1)}y = \frac{(x-1)^2}{x}(x-1)$$

Since μ is an integrating factor, the left-hand side becomes

$$\frac{d}{dx} \left(\frac{(x-1)^2}{x}y \right)$$

So

$$\frac{d}{dx} \left(\frac{(x-1)^2}{x}y \right) = \frac{(x-1)^3}{x}$$

Integrating both sides:

$$\frac{(x-1)^2}{x}y = \int \frac{(x-1)^3}{x} dx$$

Expand the integrand:

$$\begin{aligned} \frac{(x-1)^3}{x} &= \frac{x^3 - 3x^2 + 3x - 1}{x} \\ &= x^2 - 3x + 3 - \frac{1}{x} \end{aligned}$$

Hence

$$\begin{aligned} \frac{(x-1)^2}{x}y &= \int \left(x^2 - 3x + 3 - \frac{1}{x} \right) dx \\ &= \frac{x^3}{3} - \frac{3x^2}{2} + 3x - \ln x + C \end{aligned}$$

So

$$y = \frac{x}{(x-1)^2} \left(\frac{x^3}{3} - \frac{3x^2}{2} + 3x - \ln x + C \right)$$

Now use the condition $y = 2$ when $x = 2$:

$$\begin{aligned} 2 &= \frac{2}{(2-1)^2} \left(\frac{2^3}{3} - \frac{3(2^2)}{2} + 3(2) - \ln 2 + C \right) \\ 2 &= 2 \left(\frac{8}{3} - 6 + 6 - \ln 2 + C \right) \\ 2 &= 2 \left(\frac{8}{3} - \ln 2 + C \right) \end{aligned}$$

Divide by 2:

$$1 = \frac{8}{3} - \ln 2 + C$$

So

$$C = 1 - \frac{8}{3} + \ln 2 = \ln 2 - \frac{5}{3}$$

Substituting back:

$$y = \frac{x}{(x-1)^2} \left(\frac{x^3}{3} - \frac{3x^2}{2} + 3x - \ln x + \ln 2 - \frac{5}{3} \right)$$

7. Find the general solution of the differential equation

$$\frac{dy}{dx} + \cot x y = \frac{\cos x}{\sin^2 x (1 + \ln^2(\sin x))}$$

where $0 < x < \pi$

[7]

Solution

This is a first-order linear differential equation:

$$\frac{dy}{dx} + \cot x y = \frac{\cos x}{\sin^2 x (1 + \ln^2(\sin x))}$$

Its integrating factor is

$$\begin{aligned}\mu(x) &= e^{\int \cot x \, dx} \\ &= e^{\int \frac{\cos x}{\sin x} \, dx} \\ &= e^{\ln(\sin x)}\end{aligned}$$

Since $0 < x < \pi$, we have $\sin x > 0$, so

$$\mu(x) = \sin x$$

Multiply the whole differential equation by $\sin x$:

$$\begin{aligned}\sin x \frac{dy}{dx} + \sin x \cot x y &= \sin x \cdot \frac{\cos x}{\sin^2 x (1 + \ln^2(\sin x))} \\ \sin x \frac{dy}{dx} + \cos x y &= \frac{\cos x}{\sin x (1 + \ln^2(\sin x))}\end{aligned}$$

Now

$$\frac{d}{dx}(y \sin x) = \sin x \frac{dy}{dx} + \cos x y$$

So the equation becomes

$$\frac{d}{dx}(y \sin x) = \frac{\cos x}{\sin x (1 + \ln^2(\sin x))}$$

Integrating both sides gives

$$\int \frac{d}{dx}(y \sin x) \, dx = \int \frac{\cos x}{\sin x (1 + \ln^2(\sin x))} \, dx$$

For the right-hand side, let

$$u = \ln(\sin x)$$

Then

$$\begin{aligned}\frac{du}{dx} &= \frac{1}{\sin x} \cdot \cos x \\ &= \frac{\cos x}{\sin x}\end{aligned}$$

Hence

$$du = \frac{\cos x}{\sin x} \, dx$$

So

$$\begin{aligned}\int \frac{\cos x}{\sin x (1 + \ln^2(\sin x))} \, dx &= \int \frac{1}{1 + u^2} \, du \\ &= \arctan(u) + C \\ &= \arctan(\ln(\sin x)) + C\end{aligned}$$

Therefore

$$y \sin x = \arctan(\ln(\sin x)) + C$$

Finally, divide by $\sin x$:

$$y = \frac{\arctan(\ln(\sin x)) + C}{\sin x}$$

The general solution is $y = \frac{\arctan(\ln(\sin x)) + C}{\sin x}$.

8. (a) Solve the differential equation

$$(1+x) \frac{dy}{dx} - 2y = (1+x)^4, \quad x > -1$$

giving your answer in the form $y = f(x)$ [3]

(b) The solution curve passes through the point $(0, -1)$. Find the exact positive value of x for which $y = 0$ [3]

Solution

(a) Rewrite the differential equation in linear form by dividing through by $1+x$:

$$\frac{dy}{dx} - \frac{2}{1+x}y = (1+x)^3$$

Since this is a linear equation, the integrating factor is

$$\begin{aligned} \mu(x) &= e^{\int -\frac{2}{1+x} dx} \\ &= e^{-2 \ln(1+x)} \\ &= (1+x)^{-2} \end{aligned}$$

Multiplying the differential equation by $(1+x)^{-2}$ gives

$$(1+x)^{-2} \frac{dy}{dx} - 2(1+x)^{-3}y = (1+x)$$

The left-hand side is the derivative of $y(1+x)^{-2}$, so

$$\frac{d}{dx}(y(1+x)^{-2}) = 1+x$$

Integrating both sides:

$$\begin{aligned} y(1+x)^{-2} &= \int (1+x) dx \\ &= x + \frac{x^2}{2} + C \end{aligned}$$

Hence the general solution is

$$y = (1+x)^2 \left(\frac{x^2}{2} + x + C \right)$$

(b) The solution curve passes through $(0, -1)$, so substituting into the general solution from part (a):

$$\begin{aligned} -1 &= (1+0)^2 \left(\frac{0^2}{2} + 0 + C \right) \\ -1 &= C \end{aligned}$$

Therefore

$$y = (1+x)^2 \left(\frac{x^2}{2} + x - 1 \right)$$

To find when $y = 0$, set this equal to zero:

$$(1+x)^2 \left(\frac{x^2}{2} + x - 1 \right) = 0$$

Since $x > -1$, we have $1 + x > 0$, so $(1 + x)^2 \neq 0$. Therefore

$$\frac{x^2}{2} + x - 1 = 0$$

Multiply by 2:

$$x^2 + 2x - 2 = 0$$

Solve using the quadratic formula:

$$\begin{aligned} x &= \frac{-2 \pm \sqrt{2^2 - 4(1)(-2)}}{2} \\ &= \frac{-2 \pm \sqrt{4 + 8}}{2} \\ &= \frac{-2 \pm \sqrt{12}}{2} \\ &= \frac{-2 \pm 2\sqrt{3}}{2} \\ &= -1 \pm \sqrt{3} \end{aligned}$$

The positive value is

$$x = \sqrt{3} - 1$$

9. (a) Show that

$$\frac{d}{dx} \ln(\sec x + \tan x) = \sec x \quad [2]$$

(b) Hence find the general solution of the differential equation

$$\frac{dy}{dx} - 2y \tan x = \sec^3 x$$

where $-\frac{\pi}{2} < x < \frac{\pi}{2}$. [5]

Solution

(a)

$$\begin{aligned} \frac{d}{dx} \ln(\sec x + \tan x) &= \frac{\sec x \tan x + \sec^2 x}{\sec x + \tan x} \\ &= \frac{\sec x (\tan x + \sec x)}{\sec x + \tan x} \\ &= \sec x \end{aligned}$$

(b) The equation is linear, with

$$P(x) = -2 \tan x.$$

An integrating factor is

$$\begin{aligned} \mu(x) &= e^{\int -2 \tan x \, dx} \\ &= e^{2 \ln(\cos x)} \\ &= \cos^2 x, \end{aligned}$$

since $\cos x > 0$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

Multiplying through by $\cos^2 x$,

$$\cos^2 x \frac{dy}{dx} - 2y \cos x \sin x = \sec x.$$

The left hand side is

$$\frac{d}{dx} (y \cos^2 x),$$

so

$$\frac{d}{dx} (y \cos^2 x) = \sec x.$$

Using part (a),

$$y \cos^2 x = \ln(\sec x + \tan x) + C.$$

Hence

$$y = \sec^2 x (\ln(\sec x + \tan x) + C).$$

10. (a) Show that

$$\frac{d}{dx} \left(\ln \left(\frac{\cosh x + 1}{\sinh x} \right) \right) = -\operatorname{cosech} x \quad [3]$$

(b) Hence solve, for $x > 0$,

$$\sinh x \frac{dy}{dx} - y = 1 - \cosh x$$

given that $y = 2$ when $x = \ln 3$. Give your answer in exact form

[7]

Solution

(a) Write

$$\ln \left(\frac{\cosh x + 1}{\sinh x} \right) = \ln(\cosh x + 1) - \ln(\sinh x)$$

Differentiate term by term:

$$\begin{aligned} \frac{d}{dx} \left(\ln \left(\frac{\cosh x + 1}{\sinh x} \right) \right) &= \frac{\sinh x}{\cosh x + 1} - \frac{\cosh x}{\sinh x} \\ &= \frac{\sinh^2 x - \cosh x(\cosh x + 1)}{\sinh x(\cosh x + 1)} \end{aligned}$$

Now simplify the numerator using $\cosh^2 x - \sinh^2 x = 1$:

$$\begin{aligned} \sinh^2 x - \cosh x(\cosh x + 1) &= \sinh^2 x - \cosh^2 x - \cosh x \\ &= -(\cosh^2 x - \sinh^2 x) - \cosh x \\ &= -1 - \cosh x \\ &= -(\cosh x + 1) \end{aligned}$$

So

$$\begin{aligned} \frac{d}{dx} \left(\ln \left(\frac{\cosh x + 1}{\sinh x} \right) \right) &= \frac{-(\cosh x + 1)}{\sinh x(\cosh x + 1)} \\ &= -\frac{1}{\sinh x} \\ &= -\operatorname{cosech} x \end{aligned}$$

Hence,

$$\frac{d}{dx} \left(\ln \left(\frac{\cosh x + 1}{\sinh x} \right) \right) = -\operatorname{cosech} x$$

(b) Start with

$$\sinh x \frac{dy}{dx} - y = 1 - \cosh x$$

Since $x > 0$, we have $\sinh x \neq 0$, so divide through by $\sinh x$:

$$\frac{dy}{dx} - \operatorname{cosech} x y = \frac{1 - \cosh x}{\sinh x}$$

This is a linear differential equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

with

$$P(x) = -\operatorname{cosech} x$$

From part (a),

$$\frac{d}{dx} \left(\ln \left(\frac{\cosh x + 1}{\sinh x} \right) \right) = -\operatorname{cosech} x$$

so an integrating factor is

$$\mu(x) = \exp \left(\int -\operatorname{cosech} x \, dx \right) = \frac{\cosh x + 1}{\sinh x}$$

Multiply the differential equation throughout by $\mu(x)$:

$$\frac{\cosh x + 1}{\sinh x} \frac{dy}{dx} - \frac{\cosh x + 1}{\sinh^2 x} y = \frac{(\cosh x + 1)(1 - \cosh x)}{\sinh^2 x}$$

The left-hand side is

$$\frac{d}{dx} (\mu y) = \frac{d}{dx} \left(\frac{\cosh x + 1}{\sinh x} y \right)$$

Simplify the right-hand side:

$$\begin{aligned} \frac{(\cosh x + 1)(1 - \cosh x)}{\sinh^2 x} &= \frac{1 - \cosh^2 x}{\sinh^2 x} \\ &= \frac{-\sinh^2 x}{\sinh^2 x} \\ &= -1 \end{aligned}$$

So

$$\frac{d}{dx} \left(\frac{\cosh x + 1}{\sinh x} y \right) = -1$$

Integrate with respect to x :

$$\frac{\cosh x + 1}{\sinh x} y = -x + C$$

Hence

$$y = \frac{(C - x) \sinh x}{\cosh x + 1}$$

Now use the condition $y = 2$ when $x = \ln 3$.

First find $\sinh(\ln 3)$ and $\cosh(\ln 3)$:

$$\begin{aligned} \sinh(\ln 3) &= \frac{e^{\ln 3} - e^{-\ln 3}}{2} = \frac{3 - \frac{1}{3}}{2} = \frac{4}{3} \\ \cosh(\ln 3) &= \frac{e^{\ln 3} + e^{-\ln 3}}{2} = \frac{3 + \frac{1}{3}}{2} = \frac{5}{3} \end{aligned}$$

Substitute into the solution:

$$\begin{aligned} 2 &= \frac{(C - \ln 3) \sinh(\ln 3)}{\cosh(\ln 3) + 1} \\ &= \frac{(C - \ln 3) \cdot \frac{4}{3}}{\frac{5}{3} + 1} \\ &= \frac{(C - \ln 3) \cdot \frac{4}{3}}{\frac{8}{3}} \\ &= \frac{C - \ln 3}{2} \end{aligned}$$

Therefore

$$\begin{aligned} 2 &= \frac{C - \ln 3}{2} \\ 4 &= C - \ln 3 \\ C &= 4 + \ln 3 \end{aligned}$$

Substituting back,

$$y = \frac{(4 + \ln 3 - x) \sinh x}{\cosh x + 1}$$

11. (a) Find the general solution of the differential equation

$$\frac{dy}{dx} + \frac{2x}{x^2 + 1}y = 6x$$

giving your answer in the form $y = f(x)$

[3]

- (b) Given that $y = \frac{7}{2}$ when $x = 0$, find the smallest positive value of x for which $y = 4$

[3]

Solution

- (a) The differential equation is linear:

$$\frac{dy}{dx} + \frac{2x}{x^2 + 1}y = 6x$$

Its integrating factor is

$$\begin{aligned}\mu(x) &= e^{\int \frac{2x}{x^2+1} dx} \\ &= e^{\ln(x^2+1)} \\ &= x^2 + 1\end{aligned}$$

Multiplying the differential equation by $x^2 + 1$ gives

$$(x^2 + 1)\frac{dy}{dx} + 2xy = 6x(x^2 + 1)$$

The left-hand side is the derivative of $(x^2 + 1)y$, so

$$\frac{d}{dx}((x^2 + 1)y) = 6x(x^2 + 1)$$

Integrating both sides:

$$\begin{aligned}(x^2 + 1)y &= \int 6x(x^2 + 1) dx \\ &= \int (6x^3 + 6x) dx \\ &= \frac{3}{2}x^4 + 3x^2 + C\end{aligned}$$

Hence

$$y = \frac{\frac{3}{2}x^4 + 3x^2 + C}{x^2 + 1}$$

Therefore, the general solution is

$$y = \frac{\frac{3}{2}x^4 + 3x^2 + C}{x^2 + 1}$$

- (b) Using $y = \frac{7}{2}$ when $x = 0$:

$$\begin{aligned}\frac{7}{2} &= \frac{\frac{3}{2}(0)^4 + 3(0)^2 + C}{(0)^2 + 1} \\ \frac{7}{2} &= C\end{aligned}$$

So the particular solution is

$$y = \frac{\frac{3}{2}x^4 + 3x^2 + \frac{7}{2}}{x^2 + 1}$$

Now set $y = 4$:

$$4 = \frac{\frac{3}{2}x^4 + 3x^2 + \frac{7}{2}}{x^2 + 1}$$

Multiply through by $x^2 + 1$:

$$4(x^2 + 1) = \frac{3}{2}x^4 + 3x^2 + \frac{7}{2}$$

Multiply by 2:

$$8x^2 + 8 = 3x^4 + 6x^2 + 7$$

Rearranging:

$$0 = 3x^4 - 2x^2 - 1$$

Let $u = x^2$. Then

$$\begin{aligned} 3u^2 - 2u - 1 &= 0 \\ (3u + 1)(u - 1) &= 0 \end{aligned}$$

So

$$u = 1 \quad \text{or} \quad u = -\frac{1}{3}$$

Since $u = x^2 \geq 0$, we take $x^2 = 1$, giving

$$x = \pm 1$$

The smallest positive value is

$$x = 1$$

12. For $x > 0$, solve the differential equation

$$\sinh x \frac{dy}{dx} - 2y \cosh x = \sinh^2 x \cosh x$$

given that $y = 5$ when $x = \ln(1 + \sqrt{2})$.

Give your answer in exact form.

[7]

Solution

Since $x > 0$, we have $\sinh x > 0$, so in particular $\sinh x \neq 0$. We may therefore divide the differential equation by $\sinh x$ to put it into linear form:

$$\begin{aligned} \sinh x \frac{dy}{dx} - 2y \cosh x &= \sinh^2 x \cosh x \\ \frac{dy}{dx} - 2 \frac{\cosh x}{\sinh x} y &= \sinh x \cosh x \\ \frac{dy}{dx} - 2 \coth x y &= \sinh x \cosh x \end{aligned}$$

This is a first-order linear equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

with

$$P(x) = -2 \coth x$$

So the integrating factor is

$$\begin{aligned} \mu(x) &= e^{\int -2 \coth x \, dx} \\ &= e^{-2 \int \coth x \, dx} \\ &= e^{-2 \ln(\sinh x)} \\ &= \frac{1}{\sinh^2 x} \end{aligned}$$

Multiplying the differential equation by this integrating factor gives

$$\begin{aligned} \frac{1}{\sinh^2 x} \frac{dy}{dx} - \frac{2 \coth x}{\sinh^2 x} y &= \frac{\sinh x \cosh x}{\sinh^2 x} \\ \frac{d}{dx} \left(\frac{y}{\sinh^2 x} \right) &= \coth x \end{aligned}$$

Now integrate both sides with respect to x :

$$\begin{aligned} \frac{y}{\sinh^2 x} &= \int \coth x \, dx \\ &= \ln(\sinh x) + C \end{aligned}$$

Hence

$$y = \sinh^2 x (\ln(\sinh x) + C)$$

Now use the condition $y = 5$ when $x = \ln(1 + \sqrt{2})$.

We need $\sinh(\ln(1 + \sqrt{2}))$. Using

$$\sinh t = \frac{e^t - e^{-t}}{2}$$

with $t = \ln(1 + \sqrt{2})$, we get

$$\begin{aligned}\sinh(\ln(1 + \sqrt{2})) &= \frac{(1 + \sqrt{2}) - \frac{1}{1 + \sqrt{2}}}{2} \\ &= \frac{(1 + \sqrt{2}) - (\sqrt{2} - 1)}{2} \\ &= \frac{2}{2} \\ &= 1\end{aligned}$$

So substituting into the solution:

$$\begin{aligned}5 &= 1^2(\ln 1 + C) \\ 5 &= 0 + C \\ C &= 5\end{aligned}$$

Therefore the required exact solution is

$$y = \sinh^2 x (\ln(\sinh x) + 5)$$

13. For $x > 0$, a function y satisfies

$$\sinh x \frac{dy}{dx} - y \cosh x = \sinh x \cosh^3 x$$

(a) By first writing the equation in the form $\frac{dy}{dx} + P(x)y = Q(x)$, find an integrating factor [2]

(b) Hence solve the differential equation, given that $y = 2$ when $x = \ln(1 + \sqrt{2})$ [5]

Give your answer in exact form.

Solution

(a) First divide the differential equation by $\sinh x$:

$$\frac{dy}{dx} - \frac{\cosh x}{\sinh x} y = \cosh^3 x$$

So it is in the form $\frac{dy}{dx} + P(x)y = Q(x)$ with

$$P(x) = -\coth x$$

The integrating factor is

$$\begin{aligned} \mu(x) &= e^{\int P(x) dx} \\ &= e^{\int -\coth x dx} \\ &= e^{-\ln(\sinh x)} \\ &= \frac{1}{\sinh x} \end{aligned}$$

since $x > 0$, so $\sinh x > 0$.

Final answer: the integrating factor is $\frac{1}{\sinh x}$.

(b) Multiply the equation

$$\frac{dy}{dx} - \coth x y = \cosh^3 x$$

by the integrating factor $\frac{1}{\sinh x}$:

$$\frac{1}{\sinh x} \frac{dy}{dx} - \frac{\cosh x}{\sinh^2 x} y = \frac{\cosh^3 x}{\sinh x}$$

Now

$$\frac{d}{dx} \left(\frac{y}{\sinh x} \right) = \frac{1}{\sinh x} \frac{dy}{dx} - \frac{\cosh x}{\sinh^2 x} y$$

so the equation becomes

$$\frac{d}{dx} \left(\frac{y}{\sinh x} \right) = \frac{\cosh^3 x}{\sinh x}$$

Integrating both sides gives

$$\frac{y}{\sinh x} = \int \frac{\cosh^3 x}{\sinh x} dx + C$$

Rewrite the integrand using $\cosh^2 x = 1 + \sinh^2 x$:

$$\begin{aligned} \frac{\cosh^3 x}{\sinh x} &= \frac{\cosh x \cosh^2 x}{\sinh x} \\ &= \frac{\cosh x (1 + \sinh^2 x)}{\sinh x} \end{aligned}$$

Let

$$u = \sinh x \quad \Rightarrow \quad du = \cosh x \, dx$$

Then

$$\begin{aligned} \int \frac{\cosh^3 x}{\sinh x} \, dx &= \int \frac{\cosh x (1 + \sinh^2 x)}{\sinh x} \, dx \\ &= \int \left(\frac{1}{u} + u \right) du \\ &= \ln u + \frac{u^2}{2} + C \\ &= \ln(\sinh x) + \frac{1}{2} \sinh^2 x + C \end{aligned}$$

Hence

$$\frac{y}{\sinh x} = \ln(\sinh x) + \frac{1}{2} \sinh^2 x + C$$

and so

$$y = \sinh x \ln(\sinh x) + \frac{1}{2} \sinh^3 x + C \sinh x$$

Use the condition $y = 2$ when $x = \ln(1 + \sqrt{2})$. First find $\sinh(\ln(1 + \sqrt{2}))$:

$$\begin{aligned} \sinh(\ln(1 + \sqrt{2})) &= \frac{e^{\ln(1+\sqrt{2})} - e^{-\ln(1+\sqrt{2})}}{2} \\ &= \frac{(1 + \sqrt{2}) - \frac{1}{1 + \sqrt{2}}}{2} \\ &= \frac{(1 + \sqrt{2}) - (\sqrt{2} - 1)}{2} \\ &= 1 \end{aligned}$$

So substituting $\sinh x = 1$ and $y = 2$:

$$\begin{aligned} 2 &= 1 \cdot \ln 1 + \frac{1}{2}(1)^3 + C(1) \\ 2 &= 0 + \frac{1}{2} + C \\ C &= \frac{3}{2} \end{aligned}$$

Therefore

$$y = \sinh x \ln(\sinh x) + \frac{1}{2} \sinh^3 x + \frac{3}{2} \sinh x$$

14. For $x > 0$, solve the differential equation

$$\sinh x \frac{dy}{dx} - y \cosh x = \sinh x \cosh x$$

given that $y = 2$ when $x = \ln(1 + \sqrt{2})$.

Give your answer in exact form.

[7]

Solution

Since $x > 0$, we have $\sinh x > 0$, so we may divide through by $\sinh x$ to put the differential equation into linear form:

$$\begin{aligned} \sinh x \frac{dy}{dx} - y \cosh x &= \sinh x \cosh x \\ \frac{dy}{dx} - \frac{\cosh x}{\sinh x} y &= \cosh x \\ \frac{dy}{dx} - \coth x y &= \cosh x \end{aligned}$$

This is a first-order linear equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

with

$$P(x) = -\coth x$$

So the integrating factor is

$$\begin{aligned} \mu(x) &= e^{\int -\coth x \, dx} \\ &= e^{-\int \coth x \, dx} \\ &= e^{-\ln(\sinh x)} \\ &= \frac{1}{\sinh x} \end{aligned}$$

Multiplying the differential equation by this integrating factor gives

$$\frac{1}{\sinh x} \frac{dy}{dx} - \frac{\coth x}{\sinh x} y = \frac{\cosh x}{\sinh x}$$

The left-hand side is the derivative of $\frac{y}{\sinh x}$, so

$$\frac{d}{dx} \left(\frac{y}{\sinh x} \right) = \coth x$$

Integrating both sides with respect to x :

$$\begin{aligned} \frac{y}{\sinh x} &= \int \coth x \, dx + C \\ &= \ln(\sinh x) + C \end{aligned}$$

Hence

$$y = \sinh x (\ln(\sinh x) + C)$$

Now use the condition $y = 2$ when $x = \ln(1 + \sqrt{2})$.

We need $\sinh(\ln(1 + \sqrt{2}))$. Using

$$\sinh u = \frac{e^u - e^{-u}}{2}$$

with $u = \ln(1 + \sqrt{2})$:

$$\sinh(\ln(1 + \sqrt{2})) = \frac{(1 + \sqrt{2}) - \frac{1}{1 + \sqrt{2}}}{2}$$

Now rationalise:

$$\begin{aligned}\frac{1}{1 + \sqrt{2}} &= \frac{1 - \sqrt{2}}{(1 + \sqrt{2})(1 - \sqrt{2})} \\ &= \frac{1 - \sqrt{2}}{1 - 2} \\ &= \sqrt{2} - 1\end{aligned}$$

So

$$\begin{aligned}\sinh(\ln(1 + \sqrt{2})) &= \frac{(1 + \sqrt{2}) - (\sqrt{2} - 1)}{2} \\ &= \frac{2}{2} \\ &= 1\end{aligned}$$

Substituting into the general solution:

$$\begin{aligned}2 &= 1(\ln 1 + C) \\ 2 &= 0 + C \\ C &= 2\end{aligned}$$

Therefore the exact solution is

$$y = \sinh x (\ln(\sinh x) + 2)$$

15. Find the general solution of the differential equation

$$\frac{dy}{dx} - y \tan x = \frac{1}{\sqrt{1 + \sin x}}$$

where $-\frac{\pi}{2} < x < \frac{\pi}{2}$

[7]

Solution

Write the differential equation in linear form:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

with

$$P(x) = -\tan x, \quad Q(x) = \frac{1}{\sqrt{1 + \sin x}}$$

So the integrating factor is

$$\begin{aligned} \mu(x) &= e^{\int -\tan x \, dx} \\ &= e^{\ln(\cos x)} \end{aligned}$$

Since $-\frac{\pi}{2} < x < \frac{\pi}{2}$, we have $\cos x > 0$, so

$$\mu(x) = \cos x$$

Multiply the differential equation throughout by $\cos x$:

$$\begin{aligned} \cos x \frac{dy}{dx} - y \cos x \tan x &= \frac{\cos x}{\sqrt{1 + \sin x}} \\ \cos x \frac{dy}{dx} - y \sin x &= \frac{\cos x}{\sqrt{1 + \sin x}} \end{aligned}$$

Now

$$\frac{d}{dx}(y \cos x) = \cos x \frac{dy}{dx} - y \sin x$$

So the equation becomes

$$\frac{d}{dx}(y \cos x) = \frac{\cos x}{\sqrt{1 + \sin x}}$$

Integrate both sides with respect to x :

$$y \cos x = \int \frac{\cos x}{\sqrt{1 + \sin x}} \, dx + C$$

Let

$$u = 1 + \sin x \quad \Rightarrow \quad \frac{du}{dx} = \cos x \quad \Rightarrow \quad du = \cos x \, dx$$

Then

$$\begin{aligned} \int \frac{\cos x}{\sqrt{1 + \sin x}} \, dx &= \int u^{-1/2} \, du \\ &= 2u^{1/2} \\ &= 2\sqrt{1 + \sin x} \end{aligned}$$

Hence

$$y \cos x = 2\sqrt{1 + \sin x} + C$$

Therefore

$$y = \frac{2\sqrt{1 + \sin x} + C}{\cos x}$$

So the general solution is

$$y = \frac{2\sqrt{1 + \sin x} + C}{\cos x}$$

An equivalent form is

$$y = \frac{2}{\sqrt{1 - \sin x}} + C \sec x$$